InMediaVeritas: A Decentralized News Verification Platform Using zk-SNARKs and Blockchain

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***Abstract*—The proliferation of misinformation in digital media demands trustworthy, censorship-resistant verification mechanisms that operate without relying on a single central authority for content verification. This paper presents inMediaVeritas, a blockchain-based decentralized application (DApp) designed to restore public trust through a three-tier architecture involving verified reporters, community witnesses, and news agencies. The system leverages zk-SNARKs for privacy-preserving location verification, IPFS for decentralized media storage, and on-chain consensus to establish an auditable community attestation of reported events. Beyond the core verification pipeline, we describe the companion applications that complete the system: an identity wallet for reporters and witnesses, an authority application for credential issuers, and a news agency subscription interface. We detail the system architecture, cryptographic mechanisms, incentive design, implementation experience, and key limitations.**

***Index Terms*—blockchain, zk-SNARKs, decentralized verification, misinformation, IPFS, Ethereum, digital identity, news provenance**

I. Introduction

The rapid spread of false information across digital platforms motivates verification mechanisms whose decisions and provenance can be independently audited [1]. Traditional fact-checking commonly relies on editorial organizations, platform algorithms, or designated verification services. Automated systems such as ClaimBuster can assist claim detection, but machine-learning and large-language-model approaches still face ambiguity, adversarial phrasing, and hallucination, while their outputs do not provide an immutable provenance record [2]–[4].

inMediaVeritas explores a complementary verification layer based on cryptographic provenance and community attestation. Its scope is deliberately narrower than general misinformation detection. The system records who was eligible to participate, whether a witness can privately prove proximity to the claimed event, how witnesses voted, and whether submitted media remains unchanged after submission. SHA-256 hashes and IPFS content identifiers detect post-submission modification, but the current prototype does not implement a detector for synthetic or manipulated media created before submission. Likewise, the resulting consensus is an auditable community attestation rather than a substitute for editorial judgment.

The workflow spans three actors. Credentialed reporters submit event media and metadata to a smart contract. Community witnesses near the event provide privacy-preserving location proofs and confirm or deny testimonials. After a configurable quorum is reached, the credibility-weighted result is recorded on-chain. News agencies can then retrieve verified media and its attestation record. The system therefore combines provenance, witness consensus, and privacy-preserving eligibility without requiring a single content verifier to decide the result.

This short paper contributes: (i) an end-to-end architecture integrating Ethereum-compatible smart contracts, IPFS, decentralized identity, and zk-SNARK proofs; (ii) a credibility-weighted witness workflow with quorum and slashing incentives; and (iii) a working proof of concept with mobile proof-generation and on-chain gas measurements. The contribution is primarily systems-oriented: it evaluates the feasibility of integrating existing cryptographic and decentralized technologies in a journalism workflow rather than introducing a new proving system or consensus algorithm.

II. Related Work

Automated fact-checking systems can identify or route claims for review, but they do not by themselves provide cryptographically verifiable provenance [2], [3]. Blockchain provenance systems address a different problem by anchoring hashes of digital artifacts so that later modification is detectable [5], [6]; provenance alone, however, establishes that an artifact existed in a particular form, not that the depicted event occurred as claimed.

Prior decentralized journalism projects illustrate both the potential and difficulty of blockchain-based media infrastructure. Civil combined IPFS with token-based governance but lacked the witness-verification mechanism used here and encountered adoption and token-economics challenges [7]. Po.et similarly focused on blockchain-anchored metadata and licensing rather than community verification [8]. *inMediaVeritas* adds privacy-preserving witness eligibility and an on-chain, credibility-weighted attestation layer on top of media provenance.

III. Background

Groth16 zk-SNARKs provide compact proofs with efficient on-chain verification [9]. Circom provides a language for describing arithmetic circuits for zero-knowledge applications [10], while Baby JubJub is optimized for elliptic-curve

operations inside such circuits [11]. Decentralized identifiers and verifiable credentials provide the participant identity mechanisms [12], [13]. IPFS provides content-addressed decentralized storage [14], and Kubo is its reference Go implementation.

IV. Proposed Approach

Figure 1 summarizes the interaction between reporters, witnesses, the verification contracts, decentralized storage, the identity layer, and news agencies. The blockchain stores event state, hashes, votes, and consensus outcomes, while IPFS stores the larger media objects.

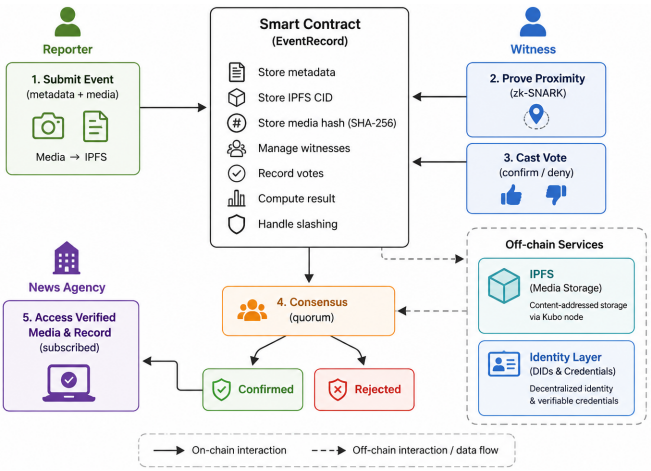


Fig. 1. Simplified architecture and verification workflow of *inMediaVeritas*.

A. Reporting and Witness Workflow

Reporters are admitted after identity verification and receive credentials linked to their decentralized identity. A submission commits an IPFS CID, timestamp, and SHA-256 integrity hash to the smart contract. The media hash and CID provide independent integrity checks: any modification after submission changes the expected fingerprint or content address. A reputation stake is associated with the submission; participation costs are pre-funded by the protocol pool.

Witness participation is open to credentialed community members who can prove that they satisfy the event's location condition. A participating witness supplies a confirm or deny testimonial together with a zk-SNARK proving that the witness location falls within the declared event radius and that the DID-linked identity commitment corresponds to a registered participant. Exact coordinates and the underlying identity are not revealed by the proof.

Accepted votes are weighted by each witness's historical credibility score, derived from alignment of previous votes with resolved consensus outcomes. The contract requires a configurable minimum quorum. If quorum is reached, the event is resolved according to the aggregate credibility-weighted outcome and recorded on-chain. If quorum is not reached within the configured time window, the event is marked unresolved and the reporter stake is returned.

News agencies form the consumer tier. The designed interface provides access to resolved events, associated media, witness-vote information, and authenticity proofs. Subscription fees are intended to fund reporter gas costs and witness rewards. The agency interface remains at the design stage, whereas the participant wallet, authority application, smart contracts, and proof circuits were implemented.

B. Identity and Cryptographic Mechanisms

The mobile identity wallet stores credentials, allows reporters to submit events and witnesses to vote, and automatically generates the required location proofs. Protocol messages are signed with keys protected by the device Secure Enclave or Keystore through biometric authentication. A separate authority application verifies applicants, issues signed credentials, and maintains an audit log. The authority is limited to participant onboarding and cannot determine content consensus, although credential issuance remains a trust dependency.

The zero-knowledge location circuit takes location and a DID-linked identity commitment as private inputs and proves both proximity and participant registration. The circuit is implemented in Circom using Groth16 over BN254 as the proving curve, with Baby JubJub used only for elliptic-curve operations embedded inside the circuit. Proofs are verified by a Solidity verifier generated from the circuit verification key. The final proof is 192 bytes and is submitted as part of the zk-enabled voting transaction.

The design therefore separates three verification properties. Credential verification determines whether a participant is admitted, the location circuit proves proximity without revealing exact coordinates, and SHA-256 together with the IPFS CID protects media integrity. None independently establishes semantic truth; the final attestation is derived from the recorded witness consensus.

C. Incentive Model

News-agency subscription fees flow into a protocol pool. Reporters receive gas reimbursement and a reward after successful resolution, witnesses receive a share based on their contribution to the resolved consensus, and stakes associated with dishonest behavior can be slashed and returned to the pool. Participation costs are pre-funded by the protocol, so users are not required to risk personal capital.

This lowers the entry barrier but limits the strength of slashing as a deterrent. The proof of concept does not provide a game-theoretic lower bound on collusion or bribery cost, nor does it establish equilibrium behavior under low agency revenue. These questions require separate adversarial and economic analysis before production deployment.

D. Security Scope and Adversarial Assumptions

The prototype considers several adversarial actions. A participant may attempt to vote without being registered, claim proximity while located elsewhere, modify submitted media after submission, or deliberately submit testimony inconsistent with other witnesses. Credential validation restricts

TABLE I
GROTH16 PROOF TIMING

Device / Circuit	Witness (ms)	Proof (ms)	Total (ms)
Pixel Identity	10.09	157.74	276.66
Pixel Location	8.46	155.09	254.31
Samsung Identity	8.90	90.65	186.53
Samsung Location	9.28	105.25	205.63
Desktop Identity	1 121	1 454	2 575
Desktop Location	951	1 411	2 362

TABLE II
SELECTED ON-CHAIN COSTS

Operation	Gas
`createEvent` (1 media)	450 615
`voteOnEvent` (first, local)	95 533
`voteOnEvent` (subsequent, local)	61 333
Identity `verifyProof`	214 787
Location `verifyProof`	227 978
zk vote, identity estimate	$\approx$ 291 000
zk vote, location estimate	$\approx$ 304 000

participation to admitted identities, the zk-SNARK circuit proves the event location condition without exposing exact coordinates, and media hashes and IPFS content identifiers make post-submission modification detectable.

These mechanisms do not prevent every form of malicious behavior. A valid participant who is physically present can still submit a dishonest testimonial, and multiple eligible participants may coordinate their votes. Credibility weighting, quorum requirements, and slashing are intended to discourage such behavior and reduce its influence, but the current prototype does not establish a bound on the number or credibility of colluding witnesses that can be tolerated. Likewise, the location proof establishes proximity rather than semantic correctness of the testimony.

The current security claim is therefore limited to verifiable eligibility, private proof of proximity, tamper-evident media provenance, and auditable consensus execution. Resistance to coordinated witness manipulation remains to be quantified through adversarial experiments and economic analysis.

V. EVALUATION

The evaluation focuses on implementation feasibility: client proof-generation latency and EVM gas consumption. Gas was measured on the Remix JavaScript VM using Solidity 0.8.13 with optimization enabled [15]. Mobile proving used *react-native-circom-witnesscalc* and *react-native-rapidsnark*; desktop proving used SnarkJS v0.7.6 under Node.js 20.17.0 on a Ryzen 5 7600X system.

Table I shows that both mobile circuits complete in under 300 ms on the tested devices. The mobile circuits use 4,505 constraints compared with approximately 7,680 in the desktop path and run through an optimized native witness/proving pipeline. The desktop values are cold-run measurements: the identity witness time of approximately 1,121 ms includes about 900 ms of Node.js startup overhead. Mobile measurements use the native *circom-witnesscalc* / *rapidsnark* pipeline, whereas the desktop measurements use the SnarkJS cold-run path. The two measurements therefore represent different proving frameworks and execution paths and should not be interpreted as a direct hardware-speed comparison between the phones and the desktop.

The two mobile devices were each measured over ten runs in the full evaluation. The Pixel 6 Pro identity and location totals were 276.66 ± 56.13 ms and 254.31 ± 8.30 ms, while the Samsung S23 Ultra totals were 186.53 ± 32.06 ms and 205.63 $\pm$ 26.66 ms. Witness calculation remained approximately 10 ms or less on the tested mobile devices; proof generation was the dominant client-side cost. Circuit setup is performed only once per circuit; subsequent proofs require only witness calculation and proving.

Creating an event with one media item consumed 450,615 gas. In the full measurements, `createEvent` scaled approximately linearly with attached media because the contract stores media metadata on-chain. A first local event vote consumed 95,533 gas, while a subsequent local vote consumed 61,333 gas because already initialized storage slots are cheaper to update. Direct Groth16 verification consumed 214,787 gas for the identity circuit and 227,978 gas for the location circuit. Combined with the measured voting-state update, an end-to-end zk-enabled media vote is approximately 291k to 304k gas, with proof verification accounting for roughly 74–75% of the cost. At the paper's reference gas price of 10 gwei, this corresponds to approximately 0.0029 ETH for an identity-based zk vote and 0.0030 ETH for a location-based zk vote.

The full gas measurements also show the storage effect explicitly. `createEvent` consumed 280,774 gas with no media, 450,615 gas with one media item, and 735,659 gas with three media items. For voting, the first write to a counter is more expensive than an update to an already nonzero counter. The locality flag can also add another storage update when the separate local tally is initialized. These effects explain differences among the non-zk vote measurements and are separate from the cryptographic verification cost. In the zk-enabled path, Groth16 verification is therefore the dominant component rather than ordinary state updates.

These measurements establish that the implemented cryptographic workflow is computationally feasible on the tested mobile devices and quantify its on-chain cost. They do not measure whether the witness consensus is correct under malicious coordination, sparse participation, false but locally consistent testimony, or compromised credential authorities. No verification-accuracy or adversarial-robustness benchmark is claimed by this evaluation.

VI. DISCUSSION AND LIMITATIONS

The main deployment limitation is witness coverage. Correct resolution requires enough geographically proximate users to participate; rural areas, unusual times, or rapidly developing events may fail to reach quorum. The explicit unresolved state prevents the system from presenting insufficiently witnessed events as verified, but it does not solve availability.

Credential issuance is another centralization risk. Multiple authorities can operate in parallel, but a malicious or captured authority could reject legitimate users or admit Sybil identities. The media relay is not a content trust anchor, yet it remains an availability dependency until clients connect directly to public or embedded IPFS nodes. Long-term media availability also depends on pinning; the full design discusses incentive-backed storage such as Filecoin [16] or permanent-storage approaches such as Arweave [17] as possible extensions. High-volume events would stress both on-chain gas usage and media pinning infrastructure, motivating L2 deployment and distributed pinning.

Legal and regulatory questions are outside the prototype evaluation. A production deployment would still need to address media licensing, copyright, data protection, and liability requirements in the jurisdictions in which agencies and participants operate.

The incentive mechanism is currently a design rather than a validated security result. Because slashing affects pre-funded protocol stakes, its deterrent value depends on reputation and future rewards, and low subscription revenue may weaken participation incentives. Similarly, credibility weighting and quorum reduce indiscriminate voting but do not by themselves prove resistance to collusion. These limitations, together with the lack of an adversarial accuracy evaluation, define the main next steps beyond the present feasibility study.

The present evaluation also does not establish domain adoption. The motivation is based on the architectural benefits of auditable provenance and distributed attestation, but the paper does not include an empirical study of newsroom trust relationships or documented cases in which agencies retracted reports because of reporter misconduct. Consequently, the prototype demonstrates a technical mechanism that agencies could use as an additional verification layer, not evidence that it should replace established editorial and correspondent relationships.

A further boundary is the distinction between event attestation and media-forensics detection. A maliciously generated artifact that is submitted in its final form will retain a consistent SHA-256 hash and CID. The integrity mechanism can show that the artifact was not changed after submission, while the local witness process is intended to attest the reported event. Detecting deepfakes or other synthetic manipulation before submission would require an additional forensic mechanism outside the current prototype.

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VII. Conclusion

inMediaVeritas demonstrates the technical feasibility of combining mobile zk-SNARK proofs, blockchain-attested witness consensus, decentralized identity, and IPFS media integrity in an end-to-end journalism prototype. The implemented system can generate proofs on the tested phones in under 300 ms and verify them on-chain at a measurable cost. The short-paper contribution is therefore a systems proof of concept and an experimentally characterized workflow, not a complete solution to misinformation. Production use still requires stronger economic and adversarial analysis, broader witness availability, and a more decentralized credential-issuance model.

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